

Tomato Leaf Disease Recognition: Research Audit, State of the Field (2026), Datasets, and a Proposed Unified Classification–Detection Pipeline

Working research plan

Prepared for the tomato disease detection research project

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1 Executive Summary

Is the current approach doable? Yes, but as stated it is not novel and the two decks contradict each other numerically. Deck A’s DenseNet121 classification and Deck B’s YOLOv8-vs-Faster-R-CNN detection are both standard 2020–2022 work; PlantVillage tomato classification is saturated (99.4–99.96% published), and “YOLOv8 beats Faster R-CNN” is a known result, not a finding. Eleven concrete inconsistencies are listed in §2 (three different accuracies for the same model, learning rates written as “1e-4 (0.00001)”, a Grad-CAM IoU that requires lesion masks the project does not have, a “fruit disease” branch trained on ripeness data).

Where the novelty actually is. Six verified gaps in the 2024–2026 tomato literature (§5): (i) no lightweight block family evaluated as *both* classifier and detector; (ii) no label-efficiency curve for self-supervised learning on tomato; (iii) no modern anomaly-detection benchmark (PatchCore/PaDiM) on tomato; (iv) no leave-one-disease-out open-set benchmark; (v) no *quantitative* explainability evaluation (every tomato XAI paper shows heat-maps as figures only); (vi) no public tomato field dataset from Gujarat.

Datasets. §4 catalogues 40+ public sets. The critical fact: the “10 000-image” and “18 000-image” tomato sets in both decks are the *same* PlantVillage data, re-split. Real cross-domain work needs Taiwan (622 raw field images), PlantDoc, Tomato-Village (Rajasthan, has detection and multi-label variants), CCMT, FieldPlant, PlantSeg (pixel masks), plus the North-Gujarat custom set. A classifier reading only 8 background pixels of PlantVillage scores 49%; a ResNet at F1 0.987 in-domain drops to F1 0.012 on Tomato-Village.

Proposed model. TOMANET: one re-parameterisable block (partial convolution + large-kernel depthwise + coordinate attention) instantiated as TOMANET-C (GAP+FC head, 224 px) and TOMANET-D (PAFPN neck of the same block + anchor-free DFL head, 640 px), both built inside a fork of Ultralytics so that the assigner, losses, augmentation and mAP evaluation are the standard ones. Target: ≤ 5 M params, ≤ 10 GFLOPs. Baselines span everything in the original survey (AlexNet→DenseNet→ViT, Faster R-CNN, YOLOv4/v5/v8) *plus* the current state of the art (YOLOv10/11/12/26, RT-DETR, ConvNeXt, EfficientNetV2, RepViT, FastViT, StarNet).

Supervised and unsupervised. Supervised is the backbone of the work. Unsupervised is worth doing but only in three specific forms (§8): self-supervised *pre-training* to cut label cost (domain SSL on 3k images beats ImageNet-21k transfer in published work), healthy-only *anomaly detection* for unknown-disease flagging (AUROC 0.94–0.99 on comparable crops), and *semi-supervised* pseudo-labelling. Unsupervised *clustering into disease classes* does not work – published NMI is 0.27 with ImageNet features and 0.64 with the best specialised method, because clusters follow species/colour/lighting, not pathology. Fully unsupervised lab→field adaptation is also a dead end for tomato: the best published number is 50.6%, whereas ten labelled field images per disease gives +11 to +29 F1.

Metrics. §10 defines everything with formulas: macro-F1/ κ /MCC/ECE for classification, COCO mAP with AP_{small} for detection (lesions are small objects), params/GFLOPs/latency on GPU/CPU/edge, cross-dataset gap and corruption robustness, pointing game / energy-PG / deletion-insertion / saliency-IoU for XAI, AUROC/FPR95 for anomaly, NMI/ARI for clustering, plus McNemar and paired tests over ≥ 3 seeds.

Cost. ≈ 900 GPU-hours (≈ 6 weeks on one 12 GB GPU), reducible to ≈ 450 by cutting seeds and the “m” scale. Fourteen-week timeline in §11.

Honest expectation. TOMANET will probably *not* beat YOLO11/YOLOv12 in-domain, and it does not need to. The defensible claims are Pareto efficiency at ≤ 3 M parameters, smaller cross-domain degradation, measured label efficiency, unknown-disease flagging, and quantified explanations – plus a released dataset. Frame the paper that way from the first draft.

2 Audit of the Current Proposal

Two versions of the slide deck were supplied: Deck A (DenseNet121 vs. a “Sequential” CNN, image classification) and Deck B (YOLOv8 vs. Faster R-CNN with Grad-CAM, object detection on leaf, fruit and pest images). Both are *technically executable*; the problems are in experimental protocol, internal consistency, and novelty. Table 1 lists what must be fixed before any of it can be written up.

Table 1: Issues found in Deck A and Deck B, with the fix adopted in this plan.

#	Issue	Why it matters	Fix in this plan
Deck A – DenseNet121 classification			
A1	Three different accuracies for the same DenseNet121 run: 99.6% (conclusion), 99.6/99.2% train/val (results), 96.2/94% (table).	Reviewers reject papers whose numbers disagree with themselves.	One results table, generated by a script from saved predictions; every number traceable to a run ID and seed.
A2	Dataset described as “high-resolution RGB photos captured in natural environments”. The 10 000-image, 10-class set is a Kaggle re-split of PlantVillage: 256×256 lab images on uniform backgrounds.	Mis-describing the data hides the biggest weakness (lab-only images do not transfer to the field; see §3).	Describe each dataset by provenance, acquisition conditions and known biases (§4); add field datasets.
A3	Learning rates written as “1e-4 (0.00001)” and “1e-3 (0.0001)”.	Arithmetic errors in hyper-parameter tables undermine trust.	Hyper-parameters stored in a versioned YAML; table generated from it.
A4	The “Sequential” baseline (85%) is a straw man; the claim “outperformed state-of-the-art” is made without comparing to any published number.	PlantVillage-tomato classification is saturated at 99–99.9% in the literature; 96–99% is not a contribution.	Compare against the full baseline set in §9, on identical splits, with mean±std.
A5	“When the gradient vanishing issue is fixed” – ReLU inside DenseNet is the stock architecture.	Presented as a contribution; it is not.	Remove. Novelty is defined in §5.
A6	80/20 train/test with “validation accuracy” reported; no separate validation split; no repeated runs.	Model selection on the test set inflates results; a single run has no error bar.	70/15/15 stratified split with deduplication; 5 seeds (or 5-fold CV); McNemar / paired test between models.
A7	Precision 0.98, recall 0.99, F1 0.99 with accuracy 99.6% – inconsistent (macro-F1 cannot exceed both P and R).	Metric definitions unclear (micro vs macro).	Report macro-P/R/F1, kappa, MCC, per-class confusion matrix (§10).
Deck B – YOLOv8 / Faster R-CNN + Grad-CAM			

#	Issue	Why it matters	Fix in this plan
B1	“Accuracy 94%” reported for a detector; fruit accuracy is 87% in the table and 89% in the text; Grad-CAM IoU 0.40 in the table, 0.39 in the text; “NSR” never defined.	Accuracy is not a detection metric unless defined (image-level? box-level at which IoU?); inconsistent values.	Report COCO-style mAP@0.5, mAP@0.5:0.95, AP _s /AP _m /AP _l , P/R/F1 at a fixed confidence; define every XAI metric with a formula (§10).
B2	Grad-CAM applied to YOLOv8 without stating the target layer, the target score (class logit? objectness? which box?) or how gradients pass NMS.	Grad-CAM needs a scalar target; for one-stage detectors the standard practice is EigenCAM / Grad-CAM on the neck output for a chosen box. Unspecified = irreproducible.	Specify: EigenCAM and Grad-CAM++ on the P3/P4 neck features, target = class score of the selected box before NMS; report pointing game and energy-based pointing game against lesion masks.
B3	“Avg. Grad-CAM IoU 0.51” requires ground-truth lesion masks; only boxes exist.	If IoU was computed against boxes, it is a box-IoU of a thresholded heatmap, not a lesion-localisation score.	Annotate a small lesion-mask subset (200–300 images) for XAI evaluation, or use the pixel-mask datasets in §4.
B4	Pipeline text describes a “classification network with self-attention, multi-head attention and transfer learning” after YOLOv8, but no such network appears in results.	Claimed component not evaluated.	Replace with the actual custom model (§7); every named component gets an ablation row.
B5	Faster R-CNN (2015) is the only comparator.	YOLOv8 > Faster R-CNN in speed and usually mAP is a known result; not a finding.	Baselines: YOLOv5/8/10/11/12, RT-DETR, plus lightweight classifiers (§9).
B6	Fruit dataset is ripe/unripe (ripeness), yet results are described as “fruit disease detection”.	Task mismatch.	Either annotate fruit disorders (blossom-end rot, cracking, sunscald) or present the fruit branch as ripeness/maturity, not disease.
B7	Custom North-Gujarat set (3 153 “mixture” images) has no described class list, annotation protocol, camera, season, or inter-annotator agreement.	This set is the strongest novelty asset in the whole project; undocumented it cannot be published or reused.	Dataset card + annotation protocol (§11.3); release on Zenodo/Mendeley with DOI.
B8	Pre-processing says “one-hot encoded vectors for multi-class classification” for a detector; 640×640 for classification too.	Copy-paste from Deck A; detectors use box labels; classifiers normally use 224–384 px.	Task-specific pre-processing (§6).
B9	Research-gap slide lists severity, multi-disease-per-leaf, field robustness, temporal tracking, explainability – but the proposed method only addresses explainability (qualitatively).	Gap and method are disconnected.	Each claimed gap maps to a measurable experiment (Table 7).

#	Issue	Why it matters	Fix in this plan
B10	Bibliography: duplicate [21][21]; numbering differs between decks; several “research observation” cells contradict the paper described (e.g. Reis 2024 listed as KNN/SVM but described as lightweight CNN + meta-learning; Nawaz 2023 “highly effective” in Deck A, “has not proven effective” in Deck B).	Credibility.	BibTeX with DOIs; observation column rewritten from the actual abstracts.

Verdict on feasibility.

- **Doable now:** YOLOv8/Faster R-CNN detection with Grad-CAM (already run), DenseNet121 classification (already run).
- **Doable with moderate effort (2–4 months on one GPU):** the custom dual-task model of §7, the cross-domain protocol, SSL pre-training, quantitative XAI.
- **Not doable as stated:** “fruit disease” without disease-annotated fruit; “severity” without lesion masks or expert severity scores; “temporal progression” without a time-series dataset. These are dropped or scoped down explicitly in §11.
- **Where the novelty actually is:** (i) a single block family that instantiates both a classifier and a detector, trained under one recipe and evaluated under one protocol; (ii) a lab→field generalisation study across five public datasets plus the North-Gujarat field set; (iii) self-supervised pre-training on pooled unlabelled tomato images with label-efficiency curves; (iv) healthy-only anomaly detection for *unknown* disease flagging; (v) quantitative, not qualitative, explainability. No single one of these is new; the combination on tomato, with a released field dataset, is publishable.

3 State of the Field (2023–2026)

3.1 Headline facts

1. **PlantVillage-tomato classification is solved.** 2024–2026 papers report 99.4–99.96% (Eff-Swin 99.70% [33], MobilePlantViT 99.40% with 0.82 M params [34], MobileNetV3-L 99.81% [82], ViT+MoE 99.96% on the full 38-class set [32]). Differences of < 0.5 pp are within seed noise. A new architecture cannot be justified by PlantVillage accuracy alone.
2. **PlantVillage is background-biased.** A classifier that sees only 8 background pixels reaches 49% on PlantVillage (chance 2.6%) [2]. Mohanty et al. (2016): 99.35% held-out $\rightarrow$ 31.4% on external web images [1]. Recent lab→field numbers: ViT+MoE 99.96% $\rightarrow$ 68–74% on PlantDoc [32]; ResNet-50 PlantVillage→PlantDoc -67.7 pp with mean confidence still 79.8% [52]; ViT-B 99.17% $\rightarrow$ 77.27% on an Indian field subset [53]; 99.72% $\rightarrow$ 41.81% (field) [54]. Jelali (2024) reviewed 35 tomato studies that use *only* real-field data and concludes PlantVillage-trained models “perform poorly on real-world images” [45].
3. **Detection has moved to field data and YOLO/RT-DETR variants.** A fair, single-recipe benchmark of YOLOv8–v12 (all sizes) on the Tomato-Village detection variant (6 classes, 14 368 images after augmentation) gives YOLOv11x mAP@0.5 0.936 / mAP@[0.5:0.95] 0.790 and YOLOv12n 0.864 / 0.604 as the best nano model [18]. This is the number to beat, at the nano/small scale.
4. **Typical “improved YOLO” paper gains are small.** Attention/neck/loss tweaks report +1 to +4 mAP@0.5 on a single dataset, single seed, e.g. improved YOLOv8n 80.2 $\rightarrow$ 81.8 on 20 534

Hainan field images [20]; YOLO11n +1.3 on 2212 Sanya images [21]. Reviewers are increasingly sceptical of this pattern.

5. **Unified/multi-task models exist but are heavy.** PMJDM (ConvNeXt shared backbone, species classification + disease detection) is 49 M params / 146 GFLOPs [39]; multi-task disease+severity distillation [40]; LDI-NET (type/disease/severity multi-label) [41]. *No lightweight ($< 5 M$) block family evaluated as both a classifier and a detector on tomato was found.*
6. **Explainability is qualitative.** Grad-CAM / Grad-CAM++ / LIME / SHAP / Eigen-CAM appear in many tomato papers [37, 29, 35], always as figures. The only quantitative saliency-vs-lesion-mask study found is on a non-tomato dataset (DeepSHAP IoU $> 85\%$) [56]. Pointing game, energy-based pointing game and deletion/insertion curves were *not* found in any tomato paper. One 2026 paper explicitly lists lack of IoU-based heat-map scoring as a limitation [55].
7. **Foundation models are arriving.** PlantCaFo (CLIP+DINO+DINOv2 adapters, 16-shot 93.5% on 38-class PlantVillage) [78]; BLIP-2+LoRA with GroundingDINO/SAM-2 zero-shot leaf masking on tomato, 97.27%, runs on Raspberry Pi 5 [38]; DINOv2 LoRA $>$ ImageNet-supervised on four agri datasets [79]; AgriCLIP [80]; SCOLD/LeafNet 186k image-caption pairs [81]. For a single-GPU project these are *baselines and feature extractors*, not things to re-train.

3.2 Representative 2024–2026 results

Table 2: Selected recent tomato disease papers. n/r = not reported. Numbers as reported by the authors (single run unless stated).

Ref	Model	Dataset	Headline	Params / FLOPs; limitation
<i>Detection – field / mixed data</i>				
Ramos & Sappa 2025 [18]	YOLOv8-v12 benchmark	Tomato-Village det., 6 cls	v11x mAP50 0.936 / 0.790; v12n 0.864 / 0.604	YOLO only; 4×A100
Chen 2025 [20]	YOLOv8n + DynamicConv, SimAM, DySample	Hainan field, 11 cls, 20 534	mAP50 81.8 / 66.3	4.4 M / 6.9 G; unvalidated elsewhere
Meng 2025 [21]	YOLO11n + MSF neck	Sanya field, 9 cls, 2 212	mAP50 76.5 / 60.5	7.9 G, 400 FPS; weak on early lesions
Wang 2024 [22]	YOLOv6 + CBAM + BiRepGFPN (P2)	Roboflow tomato; PlantDoc; FieldPlant	93.8 / PlantDoc 77.3 / FieldPlant 66.5	P2 adds noise on FieldPlant
Sun 2025 [23]	TomatoLeafDet (YOLOv8n + P2, SMKFA)	CCMT tomato, 5 cls	mAP50 0.945 / 0.768	3.35 M / 9.8 G
Xin & Li 2024 [24]	FCHF-DETR (RT-DETR-R18 + FasterNet)	Kaggle + own, 6 cls, 3 147	mAP50 97.2 / 89.1	16.3 M / 47.8 G
Bian 2025 [25]	MSFF-DETR	LAB/ENV 7 626 each, 9 cls	97.9 (lab) / 97.0 (env)	22.3 M / 67 G
Wang & Liu 2025 [26]	TomatoGuard-YOLO (YOLOv10)	greenhouse IoT, 5 cls, 10 537	mAP50 94.2 / 72.5	2.65 MB, 130 FPS
Wang 2025 [27]	BED-YOLO (YOLOv10n + DCN + BiFPN)	65% Xinjiang field + 35% PV, 4 cls	mAP50 91.3 (base 87.4)	3.7 M / 9.8 G

Ref	Model	Dataset	Headline	Params / FLOPs; limitation
Miao 2025 [28]	SerpensGate-YOLOv8	PlantDoc, 30 cls	mAP50 64.9	5.0 M / 9.7 G; TYLCV vs leaf-spot confusion
Mostafa 2026 [29]	LeafDet (YOLOv8-derived)	PlantTom 8 cls, 7 836 (Plant-Doc+PV+Roboflow+FieldPlant)	mAP50 91.6	2.69 M / 7.3 G; Eigen-CAM qualitative
Khan 2023 [30]	TomFormer (ViT+CNN)	KUTomaDATA 939; PlantDoc-tomato; PV	mAP 87 / 81 / 83	IoU threshold not stated
2026 [31]	YOLOv11n-DeiT + genetic HPO	Taiwan (9 cls), TLDDCV (7 cls)	Taiwan mAP50 97.8 / 93.4	n/r
<i>Classification</i>				
Salman 2025 [32]	ViT + MoE head	PV 38 cls; PlantDoc	99.96%; PlantDoc 74%; PV→PlantDoc 68%	single label only
Sun 2024 [33]	Eff-Swin	PV tomato	99.70%	lab only
Hoang 2026 [34]	MobilePlantViT (7 nets compared)	PV-based tomato	99.40%, macro-F1 99.31	0.82 M / 0.60 G
Gunasekaran 2026 [35]	DeiT-Tiny router + EfficientNet-B0 experts	PV tomato 10 cls, 14 529	97.8%, macro-F1 0.972	16.3 M; Grad-CAM only
Zhang & Liu 2026 [36]	Improved Vision Mamba	PlantDoc 27 cls	92.67%, macro-F1 91.70	31.4 M; 58% of errors inter-class
Assaduzzaman 2025 [37]	XSE-TomatoNet (EffNet-B0 + SE)	Kaggle tomato 11 cls, 32 532	99.11%	XAI qualitative
Kaur 2025 [38]	BLIP-2 + LoRA + GroundingDINO + SAM-2	PV 13 143 + 2 000 greenhouse	97.27%	2.7 B base; RPi-5
<i>Multi-task / severity / segmentation</i>				
Fu 2025 [39]	PMJDM (ConvNeXt shared; cls + det heads)	26 073 imgs, 5 species, 28 dis.	mAP50 61.8	49 M / 146 G
Liu 2024 [40]	multi-task distillation (disease + 3-level severity)	AI Challenger + PlantDoc + Taiwan, 13 030	98.9% / 95.3%	single-subject images
Li 2025 [42]	EMSAM (SAM + adapters)	PV segmentation, 26 cls	Dice 0.79, IoU 0.70	590 M total
TLD46k 2025 [16]	PV + CCMT + PlantDoc merged, 46 070 imgs	TLD46k	80.7% acc cross-source	shows merged-source difficulty

3.3 Best known numbers per field dataset (targets for E2/E3)

Dataset	Best reported	Model	Ref
Tomato-Village (det., 6 cls)	mAP50 0.936 / 0.790 (fair benchmark); 0.864 / 0.604 nano	YOLOv11x; YOLOv12n	[18]
Taiwan tomato	mAP50 97.8 / 93.4; YOLOv8 baseline 77.4 on a subset	YOLOv11n-DeiT	[31]
PlantDoc (30 cls)	mAP50 77.3 / F1 75.5; 64.9	improved YOLOv6; SerpensGate	[22, 28]
PlantDoc tomato subset	mAP 81	TomFormer	[30]
PlantDoc (2020 baseline)	mAP $\approx$ 38.9	Faster R-CNN	[3]
FieldPlant	mAP 66.5	Inception-ResNet-v2	[22]
PlantDoc classification	92.67% / macro-F1 91.7	improved YOLOv6	[22]
		improved Vision Mamba	[36]

3.4 Open problems named by 2024–2026 surveys

Problem	Surveys / evidence
Lab→field domain shift, background bias	Jelali 2024 [45]; Xu 2024 [46]; Shoaib 2025 [47]; Ghimire 2026 [48]; Noyan 2022 [2]
Class imbalance, rare diseases	Ghimire 2026; Xu 2024; DM-YOLO fails on Leaf Mold / TYLCV [19]
Multiple diseases per leaf	Xu 2024; Ghimire 2026; Salman 2025 (limitation); ALMDR (apple) [43]
Severity estimation from lesion area	Xu 2024; Das 2025 [50]; severity review [51]
Small / early lesions, fine-grained similarity	Shoaib 2025; Meng 2025; 58% of PlantDoc errors are inter-class [36]
Leakage, annotation noise, no standard protocol	Xu 2024 (same leaf across splits); Ghimire 2026 (asks for P/R/F1/AUROC, specimen-level splits)
Few-shot / open-set / unknown disease	Dong 2024 (VPT max-logit AUROC 94.8% 8-shot) [70]; Dong 2025 (VLM ensemble AUROC 98.4, FPR95 7.05) [71]; open-world prototypes [73]
Edge deployment	Gunasekaran 2025 review [49]; MobileNetV3-Small 251 ms on RPi-4 [82]
Explainability tied to pathology	Ghimire 2026; no quantitative saliency metrics on tomato
Foundation models / SSL	Ghimire 2026; Shoaib 2025; Sonavane 2026 SimCLR “data > architecture” [58]; PlantCLR [57]

4 Datasets: What Exists and What to Use

4.1 The single most important fact

Most “different” tomato datasets on Kaggle are the **same** data. PlantVillage’s tomato subset (18 160 images, 10 classes, 256×256, single detached leaf on a uniform background) is re-split, augmented and re-uploaded as `kaustubhb999/tomatoleaf` (11 000), `vipoool/new-plant-diseases-dataset`, `noulam/tomato`, `ahmadzargar/...segmented`, `wellCh4n/tomato-leaf-disease-image` and others. The “10 000 images / 10 classes” set in Deck A and the “Tomato Leaf 18 000” in Deck B are both this. **Training on one and testing on another is not a cross-dataset experiment.** Real cross-domain evaluation requires genuinely independent field data (Taiwan, PlantDoc, Tomato-Village, FieldPlant, CCMT, PlantWild, SLIF-Tomato, and the North-Gujarat set).

How bad is the shift? A ResNet with in-domain PlantVillage F1 of 0.987 scores **0.012** on Tomato-Village and 0.130 on a merged tomato benchmark [17]; a classifier using only 8 background pixels of PlantVillage reaches 49% (chance 2.6%) [2].

4.2 Recommended dataset set for this project

Dataset	Role	Why	Task	Licence
PlantVillage-tomato	Source (lab)	the community reference; needed for comparability with every prior paper; deduplicate before use	cls	CC0/open
Taiwan tomato	Field (small)	622 <i>raw</i> images, complex backgrounds; augmented 4976 version must not be split randomly	cls	CC BY 4.0
PlantDoc (tomato)	Field (web)	9 tomato classes with boxes; standard lab→field target	cls + det	CC BY 4.0
Tomato-Village	Field (India)	Rajasthan fields; three variants (multiclass, multi-label , detection); contains classes absent from PlantVillage (leaf miner, spotted wilt, nutrition deficiency)	cls + multi-label + det	unstated – contact authors
CCMT (tomato slice)	Field (Ghana)	5 tomato classes, different continent; used inside TLD46k	cls	CC BY 4.0
FieldPlant (tomato)	Field (Cameroon)	pathologist-supervised boxes, dense canopy backgrounds	det	CC BY 4.0
PlantSeg (tomato)	Masks	pixel-level lesion masks in the wild – solves the XAI/severity mask problem without new annotation	seg	CC BY 4.0 (check record version)
North-Gujarat custom	Field (ours)	the contribution; unique region/varieties	cls + det + masks	to be released CC BY 4.0

Optional additions if more field diversity is wanted, all CC BY 4.0 on Mendeley: SLIF-Tomato (Sri Lanka, 890 raw, boxes), Tomato Leaf Dataset AIUB (Bangladesh, 731 raw, boxes), Daffodil IU (2 600, 11 classes incl. leaf miner and insect damage), [jttrv2w27r](#) (2 995, 7 classes incl. Mg deficiency), TOM2024 (Burkina Faso, $\approx$ 5 520 tomato images, 14 classes incl. pests and abiotic disorders), Tomato Leaf Damage Progression (8 596 images, 20 291 boxes, **3 severity levels** – the closest thing to a severity benchmark).

4.3 Full catalogue

Table 3: Publicly downloadable tomato datasets (verified 30 Aug 2026). “approx.” where the source page could not be parsed exactly.

Dataset	Year	Images	Tomato classes	Cond.	Annot.	Licence
<i>Leaf disease – classification</i>						
PlantVillage tomato	2015	18 160	10 (TYLCV 5 357 ... mosaic 373)	lab	label	CC0/open
New Plant Diseases (Aug.)	2018	$\approx$ 22k tomato	10 (= PV)	lab	label	Kaggle: orig. authors
kaustubhb999 tomatoleaf	2020	11 000	10 (= PV)	lab	label	CC0 / CC BY 4.0 (Zenodo copy)

Dataset	Year	Images	Tomato classes	Cond.	Annot.	Licence
Taiwan tomato [4]	2020	622 raw / 4 976 aug	6 (bacterial spot, black mold, gray spot, healthy, late blight, powdery mildew)	mixed/field	label	CC BY 4.0
Tomato Disease Multiple Sources	2022	32.5k	11 (PV + powdery mildew)	mixed	label	unstated
CCMT tomato [9]	2023	part of 24 881 raw	5 (healthy, leaf blight, leaf curl, septoria, verticillium wilt)	field	label	CC BY 4.0
PlantWild [7]	2024	18.5k (89 cls) / v2	≈10 tomato	wild	label + text	CC BY-NC-ND 4.0
TLD46k [16]	2026	46 070	PV+CCMT+PlantDoc merged	mixed	label	download unconfirmed
LeafNet [81]	2026	186k (97 cls, 22 crops)	tomato included	mixed	label + caption	CC BY 4.0
<i>Leaf disease – detection / segmentation</i>						
PlantDoc [3]	2020	2 598 / 8 851 boxes; ≈700 tomato	9 tomato	field/web	boxes	CC BY 4.0
PlantDoc- Tomatoes (Roboflow)	2023	686	9	field/web	polygons	CC BY 4.0
Tomato- Village [5]	2023	≈3 231 raw	≈8 incl. leaf miner, spotted wilt, nutrition deficiency	field (Ra- jasthan)	label / multi- label / boxes	unstated
FieldPlant [6]	2023	5 170 / 8 629 leaves	≈6 tomato	field (Cameroon)	boxes	CC BY 4.0
PlantSeg [8]	2024– 25	11 458 disease + 8 000 healthy	tomato among 34 species	wild	masks	CC BY 4.0
Sylhet Agri. Univ. (farukalam)	2024	640×640	7	field	boxes	public domain
sebastianpalaciob	2024	4 323 / 15 135 boxes	9	mixed	boxes	CC BY-NC-SA
vasantharank (YOLOv8)	2025	10 853	10 (PV re-annotated)	lab	boxes	CC BY-NC-SA
Sichuan greenhouse [10]	2025	1 026	3 (viral, gray mold, bacterial wilt; leaf+fruit+stem)	greenhouse	boxes	CC BY 4.0

Dataset	Year	Images	Tomato classes	Cond.	Annot.	Licence
AIUB Bangladesh [11]	2025	731 / 1 621 ann.	7	field	boxes	CC BY 4.0
SLIF-Tomato [12]	2026	890 raw / 8 934 aug	8	field (Sri Lanka)	boxes	CC BY 4.0
Damage Progression	2026	8 596 / 20 291 ann.	3 diseases $\times$ 3 severity levels	field + syn- thetic	boxes	CC BY 4.0
TomatoDet [15]	2024	>10 000	5	greenhouse	VOC boxes	partial release only
<i>Fruit / ripeness / pest</i>						
LaboroTomato	2020	804 / 9 777 inst.	6 ripeness $\times$ size	farm	COCO masks	CC BY-NC-SA 4.0
tomatOD	2020	277 / 2 418 fruit	3 ripeness	greenhouse	COCO boxes	CC BY-NC-SA 4.0
Tomato Detection (Larxel)	2020	895	1 (“tomato”)	mixed	VOC boxes	see page
TomatoPlantfactory	2022–26	520 / 9 112 inst.	green/red	plant factory	VOC+YOLO	CC BY 4.0
SAU maturity & quality	2023	2 986 raw	mature/immature, fresh/defect	lab	label	CC BY 4.0
TOM2024 [13]	2024–25	$\approx$ 5 520 tomato	14 incl. <i>Tuta absoluta</i> , blossom-end rot, sunburn, N-excess	field (Burkina Faso)	label	CC BY 4.0
8 tomato pests (Huang)	2020	609 raw / 4 263 aug	8 pest species	web/IPM	label	CC BY 4.0
<i>Tuta absoluta</i> Tanzania	2020	4 341	healthy vs infested	net-house	label	CC BY 4.0
TomatoEbola [14]	2024	326 / 784 boxes	healthy vs <i>Tuta absoluta</i>	field (Nigeria)	boxes	CC BY 4.0

Not usable / not available. AI Challenger 2018 (tomato $\approx$ 11 610 images with severity labels) – host defunct, only unofficial mirrors. PDD271 – only $\approx$ 10 sample images per class are public. KUTomaData – no public release found. IP102 has no tomato-crop branch. Plant Pathology 2020/2021 is apple only.

Per-dataset caveats to state in the paper. PlantVillage: lab bias, near-duplicates leaking across random splits (a deduplicated release exists, Mendeley 3zwdw6y4pn). PlantDoc: web-scraped, label noise (bacterial spot vs. Septoria confounded, per the original paper), extreme imbalance (spider mites $\approx$ 2 images). Taiwan: only 622 raw images – the 4 976 version is rotation/mirror augmentation, so splitting it randomly leaks. Tomato-Village: single region (Rajasthan), licence unstated – ask the authors before redistributing. FieldPlant: cassava-dominated, tomato slice small. PlantWild/PlantSeg v1: non-commercial terms, possible overlap with PlantDoc. TOM2024: extreme imbalance (*Tuta absoluta* n =19, fusarium n =18). SLIF-Tomato: 8 934 is post-augmentation of 890 raw.

4.4 Answer to “which fruit dataset”

There is no public tomato-*fruit-disease* detection dataset of useful size. LaboroTomato, tomatOD, TomatoPlantfactory and the SAU set are *ripeness/quality*, not disease. The only fruit-disorder labels found are inside TOM2024 (blossom-end rot $n=118$, sunburn $n=51$) and the Sichuan greenhouse set (gray mold on fruit). So: present the fruit branch as **ripeness/maturity detection** (defensible, data exists), and treat fruit *disorders* as a small- n pilot only if the North-Gujarat set contains them.

5 Research Gaps and Novelty Statement

5.1 Gaps that are real (verified against 2024–2026 literature)

The gap slides in both decks list mostly *true* gaps, but they are stated in a form that any 2024 paper could also claim. Below they are restated as verified, checkable absences, each with the evidence and the experiment that closes it.

Table 4: Verified gaps.

#	Gap	Evidence that it is still open	Closed by
G1	No lightweight block family evaluated as <i>both</i> classifier and detector on tomato	Existing multi-task work is heavy (PMJDM 49 M / 146 G [39]) or single-task; Ultralytics shares a backbone across tasks but nobody has studied it for tomato disease	§7, E1/E2/E8
G2	No label-efficiency curve for SSL on tomato	Every plant-SSL paper fine-tunes with 100% labels [57, 58, 59]; best tomato semi-supervised point is TOM-SSL 72.51% @10% [60]	E4
G3	No modern anomaly-detection benchmark on tomato	PatchCore/PaDiM/SimpleNet/DRAEM never reported on tomato; only potato 0.99 [63] and cherry/pepper [64]	E5
G4	No leave-one-disease-out open-set benchmark on tomato	Tomato OOD is only the easy healthy-vs-diseased setting [70, 71] or open-vocabulary detection [44]	E5
G5	No quantitative XAI on tomato	Grad-CAM/Eigen-CAM used as figures only [37, 29, 35]; pointing game / energy-PG / deletion-insertion absent from the tomato literature; one 2026 paper lists this as its own limitation [55]	E6
G6	No frozen DINOv2 k -NN / linear-probe number for tomato	cheapest possible baseline, unpublished	E4
G7	Cross-dataset generalisation reported piecemeal	Isolated numbers exist (PV→PlantDoc −67.7 pp [52]; F1 0.987→0.012 on Tomato-Village [17]) but no systematic $N \times N$ matrix over tomato datasets for both tasks	E3
G8	Unsupervised severity never validated against expert scores on tomato	k -means lesion-ratio work exists [74] without ρ_c/R^2 ; tomato severity ground truth does exist [75]	E-severity (scoped)

#	Gap	Evidence that it is still open	Closed by
G9	Region gap: no public tomato dataset from Gujarat/western India	Indian field data is Rajasthan (Tomato-Village) only	dataset release

5.2 Gaps in the decks that must be dropped or restated

- “*No temporal / progression tracking*” – true, but there *is* now a public severity-progression set (Tomato Leaf Damage Progression, 8 596 images, 3 severity levels). Without collecting a time series you cannot claim this; either use that dataset for severity *levels* (not time) or drop the claim.
- “*Research is disease-specific, single disease at a time*” – no longer true; most 2024–2026 tomato papers are 5–11 class multi-disease.
- “*Some research only distinguishes healthy vs unhealthy*” – true of a few papers, but it is not a gap anyone will credit in 2026.
- “*Models fail to generalise to other crops*” – cross-crop generalisation is a different research problem; keep the scope on tomato and say so.
- “*Pests and weeds ignored*” – partly false: *Tuta absoluta* datasets exist (TomatoEbola, Tanzania set, TOM2024, Tomato-Village leaf miner). Restate as: pest damage classes are present but severely imbalanced ($n=19$ in TOM2024).

5.3 Novelty statement (what to claim, in order)

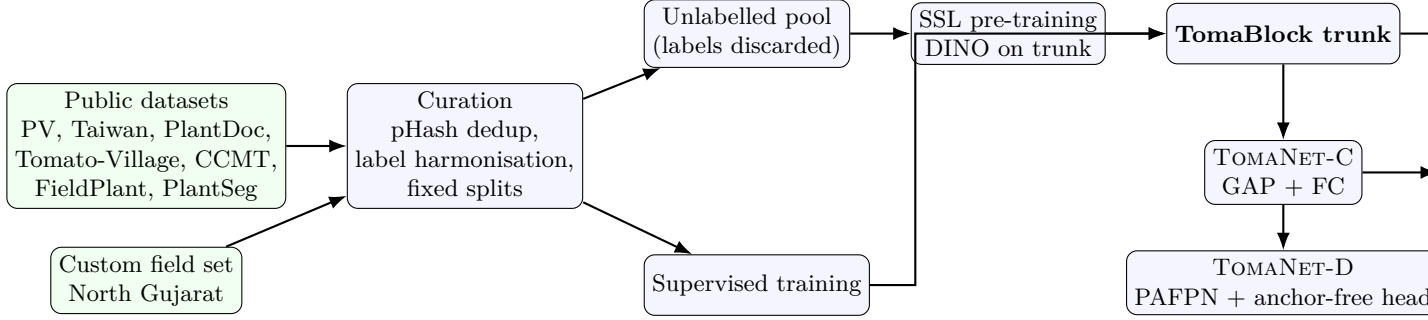
No single item below is unprecedented in machine learning; the combination, on tomato, with a released field dataset, is. Claim them in this order:

1. **A single lightweight block family (TomaBlock) instantiated as both a classifier and a detector** (TOMANET-C/TOMANET-D, ≤ 5 M params), trained under one recipe and evaluated under one protocol, with cross-task trunk transfer measured (G1).
2. **A cross-domain generalisation matrix for tomato** over ≥ 5 independent datasets for *both* tasks, reporting the gap Δ rather than only in-domain accuracy (G7).
3. **The first label-efficiency study for tomato**: SSL pre-training on a pooled unlabelled tomato corpus vs. ImageNet vs. scratch, at 1/5/10/25/100% labels, including a frozen-DINOv2 reference (G2, G6).
4. **The first unknown-disease benchmark on tomato**: healthy-only anomaly detection plus leave-one-disease-out open-set, with AUROC/FPR95 (G3, G4).
5. **The first quantitative explainability evaluation on tomato**: pointing game, energy-based pointing game, saliency-lesion IoU and deletion/insertion, with the Adebayo sanity check, against PlantSeg masks and our own polygons (G5).
6. **A documented, DOI-released North-Gujarat field dataset** with an annotation protocol and inter-annotator agreement (G9).

What *not* to claim. Do not claim state-of-the-art PlantVillage accuracy (saturated at 99.4–99.96%). Do not claim to beat YOLOv11x on Tomato-Village at equal scale without the numbers (the fair benchmark is $\text{mAP}@0.5$ 0.936 / $\text{mAP}@[0.5:0.95]$ 0.790 [18]); target the *nano* operating point, where the published reference is YOLOv12n at 0.864/0.604. Do not claim severity for non-lesion diseases (TYLCV, mosaic, mite damage). Do not claim real-time field deployment without measuring latency on the actual device.

6 Proposed Pipeline

6.1 Overview



6.2 Stage-by-stage

S1 – Curation.

Download from *original* sources (GitHub/Mendeley/Zenodo), not Kaggle re-uploads, so licences and provenance are traceable. Perceptual-hash deduplication within and *across* datasets (PlantVillage derivatives overlap heavily). Map every dataset’s class names onto the canonical vocabulary (Table 8). Emit fixed split files + a manifest with per-image SHA-256.

S2 – Task-specific pre-processing.

Classification: resize shorter side, random-resized-crop to 224 (256 variant for PlantVillage native), ImageNet normalisation, RandAugment + Mixup/CutMix. *Detection*: letterbox to 640, mosaic + HSV + flips + scale, closed mosaic for the last 10 epochs. One-hot encoding is used only for the classifier; detection uses box labels (this corrects B8).

S3 – Pre-training.

DINO on the unlabelled pool (§8), producing trunk weights that both heads can load.

S4 – Supervised training.

TOMANET-C and TOMANET-D from the shared trunk, plus all baselines under identical recipes.

S5 – Inference.

Two deployment modes: (a) *whole-image classification* for single-leaf/phone-macro use; (b) *detection* for canopy images, which yields per-lesion boxes and therefore multi-disease-per-leaf output. Both emit a calibrated confidence (temperature scaling fitted on validation) and an anomaly score; if the anomaly score exceeds the threshold the prediction is returned as *unknown* rather than forced into one of the 10 classes.

S6 – Explanation and severity.

Saliency map for the predicted class/box, scored quantitatively (§10); lesion-area ratio inside boxes for the four lesion-forming classes.

6.3 Deliverables

Code (fork of Ultralytics + a thin experiment runner), split manifests, trained weights, the North-Gujarat dataset with DOI, and one results script that regenerates every table in the paper from

saved predictions – so that no number is ever typed by hand (this is what prevents A1/B1 from recurring).

7 Proposed Custom Model: TomaNet

7.1 Design requirements

1. One block definition (TOMABLOCK) reused, without modification, in the classifier trunk, the detector trunk *and* the detector neck.
2. Two architectures built from it: TOMANET-C (image classification, 224–256 px input) and TOMANET-D (object detection, 640 px input). Same stem and stage layout, so a trunk pre-trained for one task initialises the other.
3. Budget: ≤ 5 M parameters and ≤ 10 GFLOPs at 640 px for the “n” scale, so training fits on one 8–12 GB GPU and inference runs on a phone / Jetson.
4. Every architectural choice is justified by a small-lesion or field-image argument and gets an ablation row.

7.2 TomaBlock

Input $X \in \mathbb{R}^{C \times H \times W}$, expansion ratio $e=2$, partial ratio $r=1/4$.

$$\begin{aligned}
 X_1 &= \text{PConv}_{3 \times 3}(X; r) && \text{spatial mixing on } rC \text{ channels only (FasterNet)} && (1) \\
 X_2 &= \sigma(\text{BN}(\text{Conv}_{1 \times 1}(X_1; C \rightarrow eC))) && \text{expand} && (2) \\
 X_3 &= \text{DWConv}_{k \times k}(X_2) \text{ (} k=3 \text{ in stages 1–2, 5 in stages 3–4)} && \text{large-kernel context} && (3) \\
 X_4 &= \text{CA}(X_3) && \text{coordinate attention (H- and W-wise pooling)} && (4) \\
 Y &= X + \text{BN}(\text{Conv}_{1 \times 1}(X_4; eC \rightarrow C)) && \text{project + residual} && (5)
 \end{aligned}$$

with $\sigma = \text{SiLU}$. At training time the 3×3 PConv has a parallel 1×1 branch and an identity branch (RepVGG style); the three are fused into a single 3×3 kernel for inference. Rationale:

- **PConv** keeps FLOPs and *memory access* low (the real bottleneck on edge devices), unlike depthwise-only designs.
- **Depthwise 5×5 in deep stages** gives lesion-scale receptive field growth without extra parameters (ConvNeXt finding).
- **Coordinate attention** encodes position along H and W ; lesions are small, position-specific, and easily lost by channel-only SE/ECA. CA is cheap ($\sim 2\%$ params).
- **Re-parameterisation** gives multi-branch training accuracy at single-branch inference cost.

A stage is n_i TOMABLOCKS preceded by a stride-2 3×3 conv (downsample). A TOMASTAGE(C_i, n_i) therefore has a single hyper-parameter pair per stage.

7.3 Trunk (shared)

Scale	Stem ch.	$C_1..C_4$	$n_1..n_4$	Params (trunk, approx.)	Output strides
n	16	32, 64, 128, 256	1, 2, 2, 1	≈ 1.1 M	4, 8, 16, 32
s	32	64, 128, 256, 512	1, 2, 2, 1	≈ 4.3 M	4, 8, 16, 32
m	48	96, 192, 384, 576	2, 4, 4, 2	≈ 11 M	4, 8, 16, 32

Stem: two 3×3 stride-2 convs (like YOLOv8), i.e. stride 4 before stage 1. Parameter counts are estimates from the block formula and must be replaced by `thop/fvcore` measurements once implemented (§7.7).

7.4 TomaNet-C – classification architecture

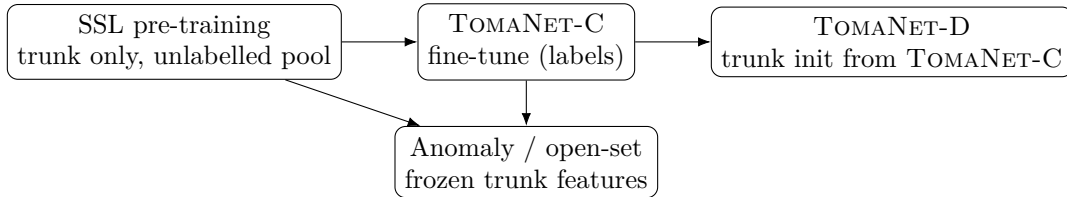
Stem $\rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow \text{Conv}_{1 \times 1}(C_4 \rightarrow 1280) \rightarrow \text{GAP} \rightarrow \text{Dropout}(0.2) \rightarrow \text{FC}(1280 \rightarrow K)$

This is exactly the Ultralytics `Classify` head (`Conv(c1,1280,1)  $\rightarrow$  AdaptiveAvgPool2d  $\rightarrow$  Dropout  $\rightarrow$  Linear`), so the YAML for TOMANET-C is the shared backbone list plus one `Classify` line – the same pattern Ultralytics uses for `yolo11.yaml` vs. `yolo11-cls.yaml`, whose backbone lists are byte-identical. Input 224 px for parity with ImageNet-pretrained baselines; 256 px variant for PlantVillage native resolution. $K = 10$ (PlantVillage tomato) or dataset-specific. The 1×1 expansion before pooling follows MobileNetV3 and costs almost nothing at 7×7 resolution. Loss: cross-entropy with label smoothing 0.1; class-balanced (effective-number) weights when the imbalance ratio exceeds 5.

7.5 TomaNet-D – detection architecture

- **Backbone:** same stem + $S_1..S_4$; export P2 (stride 4), P3 (8), P4 (16), P5 (32). SPPF on P5.
- **Neck:** PAFPN (top-down + bottom-up) in which every fusion node is a TOMABLOCK (replacing YOLOv8’s C2f). A *P2 head is optional* and ablated: lesions on a 640-px field image are frequently < 16 px, i.e. below one P3 cell.
- **Head:** anchor-free decoupled head (separate cls / reg branches) with Distribution Focal Loss (DFL, 16 bins) for box regression, exactly as in YOLOv8, so that Ultralytics’ Task-Aligned Assigner and mAP evaluation can be reused unchanged.
- **Losses:** $\mathcal{L} = \lambda_{cls} \text{BCE} + \lambda_{box} \mathcal{L}_{IoU} + \lambda_{dfl} \text{DFL}$ with default weights (0.5, 7.5, 1.5). $\mathcal{L}_{IoU} \in \{\text{CIoU}, \text{WIoU-v3}, \text{MPDIoU}\}$ is an ablation; recent small-object papers favour WIoU/MPDIoU; the plan starts from CIoU (the default) and only adopts an alternative if it wins on the validation split.
- **Post-processing:** standard NMS (IoU 0.7, conf 0.25 for reporting P/R/F1; conf 0.001 for mAP). Multi-label per leaf is handled natively: several boxes of different classes on one leaf. An NMS-free variant (YOLOv10-style dual assignment: one-to-many TAL top- $k=10$ + one-to-one top- $k=1$ head, `end2end=True` in the `Detect` head) is a cheap optional ablation because Ultralytics already implements it.
- **Loss caution:** recent agricultural papers report +0.5 to +2 mAP@0.5 from WIoU/MPDIoU/Inner-IoU/NWD, almost always single-dataset and single-seed – within typical seed variance (≈ 0.3 –0.5 mAP). Only CIoU/DIoU/GIoU are switchable in Ultralytics; others require patching `BboxLoss.forward`. Ablate with ≥ 3 seeds before claiming anything.

7.6 Weight-sharing and the transfer chain



Three initialisations of the trunk are compared for both tasks: (a) random, (b) ImageNet-1k supervised (train TOMANET-C on ImageNet-1k for 100 epochs – feasible at scale “n” in ≈ 2 GPU-days at 160 px; if not affordable, use ImageNet-100), (c) self-supervised on the pooled tomato images (§8). This is the cleanest way to make the “unsupervised” track a measurable part of the main results rather than a side experiment.

7.7 Implementation route

Do not write a training loop. Both tasks are implemented inside **Ultralytics 8.x**, which already supports a `classify` task (`Classify` head with GAP + FC) and a `detect` task with the TAL

assigner, DFL, mosaic augmentation, EMA, AMP, mAP evaluation and export (ONNX, TensorRT, TFLite).

1. Implement `TomaBlock`, `TomaStage`, `CoordAtt`, `PConv` in one file `ultralytics/nn/modules/toma.py`. Ultralytics has *no* plugin registry (a “Python-in-YAML” PR was closed on security grounds), so work on an editable fork (`pip install -e .`): export the classes in `nn/modules/__init__.py`, import them in `nn/tasks.py`, add them to the `base_modules` set (so c_1, c_2 are injected with width scaling) and to `repeat_modules` (so the YAML `repeats` column becomes n). Block signature must be `(c1, c2, *args)`.
2. `tomanet-cls.yaml`: backbone list + `Classify` head. `tomanet-det.yaml`: same backbone list + SPPF + PAFPN of `TomaBlock` + `Detect` head. Scales `n/s/m` through the standard `scales:` dictionary (depth, width, max-channels).
3. Verify with `model.info()` (params, GFLOPs) and one forward pass per task; unit test: fused vs unfused block outputs agree to 10^{-4} .
4. SSL pre-training uses the same trunk exported as a plain `nn.Module` into a Lightly / solo-learn DINO or MAE recipe; weights are loaded back with a key-mapping script.
5. FLOPs/latency: `thop` for GFLOPs, `torch.utils.benchmark` for GPU latency, ONNX Runtime for CPU latency, TensorRT-FP16 on Jetson if hardware is available.

7.8 Ablation list (one row each)

Abl-1 Full block vs. block without CA vs. CA replaced by SE / ECA / CBAM.

Abl-2 PConv vs. full 3×3 conv vs. depthwise-only.

Abl-3 Rep-branch on/off.

Abl-4 $k = 5$ in deep stages vs. $k = 3$ everywhere.

Abl-5 Detector: with vs. without P2 head; TOMABLOCK neck vs. C2f neck.

Abl-6 Loss: CIoU vs. WIoU-v3 vs. MPDIoU.

Abl-7 Trunk init: scratch vs. ImageNet vs. SSL (tomato pool).

Abl-8 Input resolution: 224 vs. 256 (cls); 640 vs. 800 (det).

8 Unsupervised and Self-Supervised Track

8.1 Direct answer: is unsupervised learning suitable here?

Partly. The evidence from 2021–2026 splits cleanly into four verdicts.

Approach	Verdict	Evidence
Self-supervised <i>pre-training</i> , then fine-tune with few labels	Use it	Domain SSL on only 3000 unlabelled agricultural images beats ImageNet-21k supervised transfer (+7.90 vs +5.55 over random init); gain grows with pool size: 500 imgs +1.56, 1k +2.78, 2k +4.00, 3k +4.57 [58]. PlantCLR (SimCLR + ConvNeXt-T) 99.10% PV / 96.83% field cassava [57]. Frozen DINOv2 linear probe $\approx$ 97% on leaf disease, consistently > ImageNet-1k pre-training [79].
Healthy-only <i>anomaly detection</i> for unknown disease	Use it	Colour-reconstructability (pix2pix + CIEDE2000) AUROC 0.99 on PV potato trained on 76 healthy leaves; AnoGAN 0.97, OC-SVM 0.95, AE-SSIM 0.94 [63]. VQ-VAE 0.983 (cherry) / 0.936 (pepper) [64]. Healthy-ID vs diseased-OOD on PV: AUROC 99.6–99.85, FPR95 0.71–1.68 [71].
Semi-supervised (pseudo-label, Mean Teacher, FixMatch)	Use it, modestly	Mean Teacher 88.50% with 5% labels on 38-class PV [61]; AaSSL +1.7 to +3.4pp over supervised at 5–20% labels, < 1 pp at $\geq$ 50% [62]. On tomato specifically only one weak result exists: TOM-SSL 72.51% at 10% labels (PV-tomato), 70.87% (Taiwan) [60] – a low bar.
Unsupervised <i>clustering</i> to recover disease classes	Do not	Raw ResNet-50 <i>k</i> -means on PV apple+grape (8 classes): NMI 0.272, purity 0.396. The best specialised method (CIKICS) reaches NMI 0.636 / purity 0.683 / F1 0.835 [65]; on citrus only NMI 0.331. Clusters follow species, colour and lighting, not pathology; early blight / Septoria / target spot merge. Use clustering only for pre-annotation and active-learning proposals.
Fully unsupervised domain adaptation lab→field	Not alone	Tomato is the <i>worst</i> target in the UDA literature: MSUN reaches 50.58% on tomato (vs 96.78% on corn), from a 41.81% source-only baseline [67]. Ten labelled field images per disease with metric-based few-shot DA gives +11.1 to +29.3 F1 [68] – far better than any unsupervised method. CycleGAN field→lab translation reaches F1 95.6 but only by translating field images into lab style at inference [69].

8.2 What the literature has *not* done on tomato (the openings)

These are verified absences, each cheap to fill and each a defensible contribution:

- U1 **No label-efficiency curve for any SSL method on tomato.** Every plant-SSL paper found fine-tunes with 100% of labels. A 1/5/10/25/100% curve on PV-tomato *and* on field tomato, with SSL vs ImageNet vs scratch, does not exist.
- U2 **No modern anomaly-detection benchmark on tomato.** PatchCore, PaDiM, SimpleNet, DRAEM and reverse distillation have never been reported on tomato healthy→diseased. Only potato (0.99) and cherry/pepper (0.98/0.94) exist.
- U3 **No NMI/ARI/purity on the 10-class PV-tomato subset.** CIKICS covered apple/grape/peach/strawberry; the one tomato clustering paper used 300 images, 3 classes, and reported only internal indices (silhouette 0.273) [66].
- U4 **No leave-one-disease-out open-set benchmark on tomato.** Existing tomato OOD work is the easy setting (healthy = in-distribution, any disease = OOD) or open-vocabulary detection [44]. “9 diseases known, 1 held out” with AUROC/FPR95 is unreported.
- U5 **No frozen DINOv2/v3 *k*-NN or linear-probe number on tomato,** despite it being the cheapest possible baseline.
- U6 **Unsupervised lesion-ratio severity on tomato is never validated against expert scores** with concordance ρ_c or R^2 , although tomato severity ground truth exists (bacterial spot, 150 leaves, raters ρ_c 0.83→0.96 with standard area diagrams [75]).

8.3 Track U1 – Self-supervised pre-training (feeds E4)

- **Pool:** union of all training images from every dataset in §4 plus unannotated Custom images, labels discarded ($\approx$ 40–60k images). Test images excluded by hash.

- **Method:** DINO (multi-crop $2 \times 224 + 6 \times 96$, teacher momentum $0.996 \rightarrow 1$, 300 epochs) on the TOMANET trunk. Fallback if compute-bound: SimCLR/MoCo-v3, 200 epochs. Note that [58] obtained +4.57 pp from only 3 000 images, so the pool size here is comfortable.
- **Evaluation:** k -NN ($k=20$, cosine, $T=0.07$) and linear probe on frozen features, then the label-efficiency curve of U1 at 1/5/10/25/100% labels $\times$ 3 seeds, for *both* classification and detection (detection uses the SSL trunk as initialisation).
- **Reference points to beat:** TOM-SSL 72.51% at 10% labels on PV-tomato [60]; Mean Teacher 88.50% at 5% labels on 38-class PV [61]; frozen DINOv2 linear probe (run it – it is one afternoon of compute and currently unpublished for tomato, U5).

8.4 Track U2 – Unknown-disease flagging (feeds E5)

Two settings, both trained *without* labels for the target condition:

1. **Healthy-only anomaly detection.** Memory-bank (PatchCore) over frozen TOMANET trunk features of healthy leaves only; score = distance to the bank. Compare against PaDiM, an autoencoder and the colour-reconstructability baseline [63]. Report image-level AUROC/AUPR/FPR95 on lab images *and* on field images – the expected result is a large drop on field backgrounds (pepper already shows 0.936 vs cherry 0.983), and documenting that drop is itself the finding.
2. **Leave-one-disease-out open set (U4).** Train on 9 classes, hold out 1; score with (a) max-softmax, (b) energy, (c) PatchCore distance, (d) Mahalanobis on trunk features. Rotate the held-out class over all 10; report mean AUROC/FPR95 per held-out disease. Expect max-softmax to be weak – OpenMax reached only 0.812 AUROC / 70.1% accuracy on strawberry, while outlier-exposure and energy/VPT tuning reached 0.972 and 94–98 [72, 70].

8.5 Track U3 – Severity as lesion-area ratio (scoped)

Inside each predicted box: SAM-2 (prompted with the box) for the leaf mask, then k -means in Lab colour space *inside* the mask for lesion pixels; severity = $|\text{lesion}|/|\text{leaf}|$. Validate against the 300-image polygon subset (Pearson r , MAE) and, if an agronomist can score them, against ordinal severity classes (weighted κ), which would close U6. Two documented cautions: vanilla SAM is poor at *lesions* directly (adapters add +41 Dice [77]; zero-shot leaf recall 63.2 vs 78.7 for trained Mask R-CNN [76]), so SAM is used for the *leaf* mask only; and colour-ratio severity is meaningless for TYLCV, mosaic and mite damage, which have no discrete lesions – severity is therefore reported only for the four lesion-forming classes (early blight, late blight, Septoria, target spot) and explicitly declared not applicable elsewhere.

8.6 What is deliberately *not* attempted

Fully unsupervised lab→field adaptation as a headline result (best published tomato number is 50.58%); clustering as a substitute for labels; unsupervised discovery of new disease *classes* (only flagging that something is unknown). These are stated as limitations rather than quietly omitted.

9 Baselines and Comparative Models

Every model that appears in the literature survey of the slides is included, plus the current (2025–2026) state of the art; otherwise the comparison is dated. All baselines are trained on the *same* splits with the *same* recipe family (§11) and reported with params, GFLOPs and latency. Parameter/FLOP figures below are the standard published values (ImageNet, 224 px for classifiers; 640 px for detectors) and are re-measured with `thop` in the final tables. **Convention:** Ultralytics reports GFLOPs = $2 \times \text{MACs}$ at 640 px; `thop/ptflops` return MACs; `fvcore` counts 1 MAC as 1 FLOP. The final tables state one convention and re-measure every model with the same tool. Detector mAP context on COCO (n-scale): YOLOv8n 37.3, YOLOv10n 39.5, YOLO11n 39.5, YOLOv12n 40.6, YOLO26n 40.9.

Table 5: Classification baselines.

Model	Params (M)	GFLOPs	In slides?	Role
AlexNet	61	0.7	yes	historical; Chen 2022 (modified AlexNet)
VGG16	138	15.5	yes	historical; Younis 2022, Nawaz 2023
ResNet50	25.6	4.1	yes	standard CNN reference
DenseNet121	8.0	2.9	yes	the model in Deck A; Abbas 2021 (C-GAN + DenseNet121)
InceptionV3	23.8	5.7	yes	Hosen 2025
EfficientNet-B3	12.2	1.9	yes	Ullah 2023 (EffiMob-Net), Ahmad 2021
MobileNetV2	3.5	0.3	yes	Sohib 2022; lightweight reference
MobileNetV3-Large	5.4	0.22	no	modern lightweight reference
SqueezeNet 1.1	1.2	0.35	yes	Thuseethan 2022 (Siamese), Guha Paul 2023
EfficientNetV2-S	21.5	8.4	no	strong modern CNN
ConvNeXt-Tiny	28.6	4.5	no	modern CNN; same design family as TOMANET (large-kernel DW)
ViT-B/16	86	17.6	yes	Gangwar 2023; transformer reference
Swin-Tiny	28.3	4.5	no	hierarchical transformer reference
MobileViT-S / EfficientViT-M2	5.6 / 4.2	2.0 / 0.2	no	hybrid lightweight; Thai 2025 (CNN+Transformer)
YOLOv8n-cls / YOLO11n-cls / YOLO26n-cls	2.7 / 1.6 / 2.8	0.5 / 0.5 / 0.5 (224 px)	no	same framework as TOMANET-C; fairest lightweight comparison (ImageNet top-1 $\approx$ 69 / 70 / 71.4)
RepViT-M0.9 / FastViT-T8 / StarNet-S1	5.1 / 3.6 / 2.9	0.8 / 0.7 / 0.4	no	2023–2024 mobile SOTA; closest competitors to TOMANET by design
TomaNet-C-n/s	target 1.3 / 4.6	target 0.3 / 1.0	–	proposed

Table 6: Detection baselines (640 px).

Model	Params (M)	GFLOPs	In slides?	Role
Faster R-CNN R50-FPN	41.5	$\approx$ 180	yes	two-stage reference (Deck B)
Mask R-CNN R50-FPN	44	$\approx$ 260	yes	Junare 2025; only if masks exist
YOLOv4 / YOLOv4-tiny	64 / 6	60 / 7	yes	Wang 2021, Gangwar 2023
YOLOv5s / YOLOv5n	7.2 / 1.9	16.5 / 4.5	yes	Oni 2025
YOLOv8n / s	3.2 / 11.2	8.7 / 28.6	yes	Deck B model; Wang 2023
YOLOv10n / s	2.3 / 7.2	6.7 / 21.6	no	NMS-free
YOLO11n / s	2.6 / 9.4	6.5 / 21.5	no	current Ultralytics default
YOLOv12n / s	2.6 / 9.3	7.6 / 23.7	no	attention-centric (area attention), Feb 2025

Model	Params (M)	GFLOPs	In slides?	Role
YOLO26n / s	2.4 / 9.5	5.4 / 20.7	no	Jan 2026: NMS-free end-to-end, DFL removed, small-target-aware assignment (STAL)
RT-DETR-R18	20	60	no	transformer detector reference
TomaNet-D-n/s	target 2.5 / 8	target 7 / 22	–	proposed

Fairness rules. (i) identical train/val/test image lists (hashes published); (ii) identical input resolution per task; (iii) identical epochs and early-stopping rule; (iv) baseline hyper-parameters left at their official defaults (Ultralytics for detectors, `timm` recipe for classifiers) – tuning only TOMANET would bias the comparison; (v) 3–5 seeds per model, mean±std; (vi) ImageNet initialisation for all baselines that have it, and both ImageNet and SSL initialisation for TOMANET so the effect of pre-training is isolated from the effect of architecture.

10 Evaluation Metrics

All metrics are computed by `torchmetrics` / `pycocotools` / Ultralytics validators from saved predictions; no hand-computed numbers. Notation: K classes, N test images, confusion matrix M_{ij} (true i , predicted j), TP_k, FP_k, FN_k per class.

10.1 Classification

$$\text{Accuracy} = \frac{1}{N} \sum_k M_{kk} \quad (6)$$

$$P_k = \frac{TP_k}{TP_k + FP_k}, \quad R_k = \frac{TP_k}{TP_k + FN_k}, \quad F1_k = \frac{2P_k R_k}{P_k + R_k} \quad (7)$$

$$\text{Macro-F1} = \frac{1}{K} \sum_k F1_k \quad (\text{primary metric; robust to imbalance}) \quad (8)$$

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_o = \text{Accuracy}, \quad p_e = \sum_k \frac{(\sum_j M_{kj})(\sum_i M_{ik})}{N^2} \quad (9)$$

$$\text{MCC} = \frac{N \sum_k M_{kk} - \sum_k t_k p_k}{\sqrt{(N^2 - \sum_k p_k^2)(N^2 - \sum_k t_k^2)}}, \quad t_k = \sum_j M_{kj}, \quad p_k = \sum_i M_{ik} \quad (10)$$

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} |\text{acc}(S_b) - \text{conf}(S_b)| \quad (B = 15 \text{ bins}) \quad (11)$$

Also: per-class AUROC (one-vs-rest), top-2 accuracy (relevant for visually confusable pairs such as early vs. late blight), full confusion matrix in the appendix. Report **in-domain** (same dataset) and **cross-domain** (train on A, test on B) values; the *generalisation gap* $\Delta = \text{Acc}_{\text{in}} - \text{Acc}_{\text{cross}}$ is a headline number.

10.2 Detection

COCO protocol via `pycocotools`:

$$\text{IoU}(b, \hat{b}) = \frac{|b \cap \hat{b}|}{|b \cup \hat{b}|} \quad (12)$$

$$\text{AP}_k(\tau) = \int_0^1 p_k(r; \tau) dr \quad (101\text{-point interpolation}), \quad (13)$$

$$\text{mAP@0.5} = \frac{1}{K} \sum_k \text{AP}_k(0.5), \quad \text{mAP@[0.5:0.95]} = \frac{1}{K} \sum_k \frac{1}{10} \sum_{\tau \in \{0.5, \dots, 0.95\}} \text{AP}_k(\tau) \quad (14)$$

plus $\text{AP}_s/\text{AP}_m/\text{AP}_l$ (area < 32², 32²–96², > 96² px – lesions are mostly “small”), AR@100, and P/R/F1 at conf 0.25 / IoU 0.5 for the deployment operating point. Primary metric: **mAP@[0.5:0.95]**; secondary: **AP_s**.

10.3 Efficiency

Params (M), GFLOPs at task resolution (**thop**), model size (MB, FP16 ONNX), latency (ms, batch 1, median of 200 runs after 20 warm-up) on: GPU (FP16), CPU (ONNX Runtime, 4 threads), and an edge device if available (Jetson Orin Nano / Raspberry Pi 5 / Android via TFLite). Report FPS = 1000/latency. Optional: energy per image (Jetson **tegrastats**).

10.4 Robustness

- **Cross-dataset matrix:** train on each of {PlantVillage, Taiwan, Tomato-Village, Custom}, test on all others; report macro-F1 (cls) / mAP@0.5 (det).
- **Corruption robustness** (ImageNet-C style, severities 1–5): Gaussian noise, motion blur, brightness, contrast, JPEG, fog; report mean corruption error relative to clean.
- **Background swap test:** PlantVillage leaves segmented (Otsu/SAM) and pasted on field backgrounds; accuracy drop quantifies background reliance.

10.5 Explainability (quantitative)

Given a saliency map $S \in [0, 1]^{H \times W}$ and a ground-truth lesion mask G (or the GT box when no mask exists – state which):

$$\text{Pointing game} = \frac{1}{N} \sum_n \mathbb{I}[\arg \max S_n \in G_n] \quad (15)$$

$$\text{Energy-based PG} = \frac{\sum_{(i,j) \in G} S_{ij}}{\sum_{(i,j)} S_{ij}} \quad (\text{fraction of saliency mass inside the lesion}) \quad (16)$$

$$\text{Saliency IoU}(\theta) = \frac{|\{S > \theta\} \cap G|}{|\{S > \theta\} \cup G|}, \quad \theta \text{ chosen so that } |\{S > \theta\}| = |G| \quad (17)$$

Deletion / Insertion AUC (RISE protocol): remove / insert pixels in order of saliency; AUC of score-vs-fraction (18)

Methods compared: Grad-CAM, Grad-CAM++, Score-CAM, LayerCAM (classifier, last stage) and EigenCAM / Grad-CAM on P3–P4 neck features for the detector (target = class score of the matched box, computed before NMS). Sanity check: model-parameter randomisation test (Adebayo et al.) – saliency must degrade when weights are randomised. “NSR” from Deck B is dropped unless it can be defined.

10.6 Unsupervised / self-supervised

- **SSL pre-training quality:** k -NN accuracy ($k=20$) and linear-probe accuracy on frozen features; label-efficiency curve: fine-tune with 1%, 5%, 10%, 25%, 100% of labels (3 seeds each).
- **Anomaly / unknown-disease detection:** image-level AUROC and AUPR, and FPR@95%TPR, for healthy vs. diseased and for known-vs-unknown (hold out 2 disease classes at train time).
- **Clustering:** NMI, ARI, purity of k -means ($k=K$) on frozen embeddings vs. true labels; silhouette score.
- **Severity (if lesion masks exist):** Pearson r and MAE between predicted lesion-area ratio and annotated ratio; agreement with ordinal severity classes via weighted kappa.

10.7 Statistics

Every table cell = mean $\pm$ std over ≥ 3 seeds (5 for headline models). Model-vs-model differences tested with McNemar (classification, paired on the same test images) or a paired t -test over seeds (mAP); report p -values for the headline claims. Publish the exact split files (image hashes) with the code.

11 Experimental Plan

11.1 Gap-to-experiment mapping

Table 7: Every gap claimed in the research-gap slide is tied to a measurable experiment or explicitly dropped.

Claimed gap	Experiment in this plan	Metric
Lab-only data; accuracy drop in the field	E3 cross-dataset matrix; background-swap test	Δ macro-F1, mAP gap
Class imbalance	class-balanced loss + resampling	macro-F1 vs accuracy gap, per-class F1
Multiple diseases on one leaf	ablation on Tomato-Village and Custom detection on Tomato-Village / Custom (multi-box, multi-class)	per-image multi-label F1
No lesion localisation	TOMANET-D boxes; saliency IoU	mAP, AP_s , saliency IoU
Severity not measured	scoped: lesion-area ratio inside boxes via SAM / colour clustering on a 300-image mask subset	Pearson r , MAE
Unknown / unseen disease	E5 anomaly + open-set (hold out 2 classes)	AUROC, FPR@95
Explainability qualitative only	E6 quantitative XAI	PG, energy-PG, deletion AUC
Fruit “disease”	scoped to ripeness / maturity detection on LaboroTomato + Kaggle fruit; disorders only if annotated	mAP
Temporal progression	dropped – no time-series data	–
Pests	only if the Custom set contains annotated pest classes	mAP

11.2 Data protocol

1. **Deduplicate** every dataset with perceptual hashing (pHash, Hamming ≤ 6); PlantVillage and its Kaggle derivatives contain near-duplicates and augmented copies, which leak across splits. Report how many were removed.
2. **Splits**: stratified 70/15/15 (train/val/test) per dataset, fixed by seed 0, released as file lists. Model selection on val only; test touched once per final model.
3. **Label harmonisation**: one canonical tomato class vocabulary (Table 8) with a mapping from every dataset’s class names; classes absent from a dataset are marked so cross-dataset tests only score shared classes.
4. **Unlabelled pool** for SSL: union of all training images of all datasets plus unannotated Custom images, labels discarded. No test image may enter the pool.

Table 8: Canonical class vocabulary (tomato leaf). Extend as datasets require.

ID	Canonical class	Pathogen type	Typical lesion size
0	Healthy	–	–
1	Bacterial spot (<i>Xanthomonas</i> spp.)	bacterial	small (1–3 mm)
2	Early blight (<i>Alternaria solani</i>)	fungal	medium, concentric rings
3	Late blight (<i>Phytophthora infestans</i>)	oomycete	large, water-soaked
4	Leaf mold (<i>Passalora fulva</i>)	fungal	diffuse, underside
5	Septoria leaf spot	fungal	small, many
6	Two-spotted spider mite	pest damage	stippling, whole-leaf
7	Target spot (<i>Corynespora</i>)	fungal	medium, bullseye
8	Yellow leaf curl virus (TYLCV)	viral	whole-leaf (curl, yellowing)
9	Mosaic virus (ToMV)	viral	whole-leaf mottling
10+	Powdery mildew, leaf miner, nutrient deficiency, ...	as present in field sets	–

11.3 Custom North-Gujarat dataset: what to do with it

This is the project’s most valuable asset. Minimum documentation for publication (dataset card): location and dates, crop variety, growth stage, camera/phone models, distance and lighting, number of plants/fields, class list with agronomist confirmation, annotation tool (CVAT / Label Studio / Roboflow), annotation protocol (box around each lesion cluster vs. whole leaf), number of annotators and inter-annotator agreement (box IoU ≥ 0.5 agreement rate, Cohen’s κ for class), split lists, licence (CC BY 4.0). Target: all 3 153 images with image-level labels; $\geq 1\,500$ with boxes; ≥ 300 with lesion polygons (for severity and XAI). Release on Zenodo with a DOI; cite it as a contribution.

11.4 Experiments

E1 – In-domain classification.

All Table 5 models + TOMANET-C on PlantVillage-tomato, Taiwan, Tomato-Village (cls split), Custom. 5 seeds. Expected outcome: everything $\geq 99\%$ on PlantVillage (saturated); differences appear on field sets.

E2 – In-domain detection.

Table 6 models + TOMANET-D on Tomato-Village (det), PlantDoc-tomato boxes, Custom. 3 seeds. Report AP_s separately.

E3 – Cross-domain.

Train on X , test on Y for all dataset pairs; both tasks. The key figure of the paper is the heat-map of generalisation gaps, with TOMANET vs. the best baseline.

E4 – Label efficiency / SSL.

Trunk pre-trained by DINO (or MAE) on the unlabelled pool; fine-tune with 1/5/10/25/100% labels; compare to ImageNet init and scratch. Both tasks.

E5 – Unknown-disease flagging.

Hold out 2 classes (e.g. target spot, mosaic) from training; PatchCore-style memory bank on frozen trunk features of healthy + known classes; AUROC for detecting held-out classes at test time. Compare with max-softmax, energy score, and a plain healthy-only autoencoder.

E6 – Quantitative XAI.

Saliency metrics of §10 on the lesion-mask subset for TOMANET-C, TOMANET-D, DenseNet121, YOLOv8. Sanity check with randomised weights.

E7 – Efficiency.

Latency/FPS/size on GPU, CPU, and one edge device; accuracy-vs-latency Pareto plot with all models.

E8 – Ablations.

Abl-1.8 from §7, on Tomato-Village (det) and Taiwan (cls), 3 seeds.

11.5 Training recipes (fixed across models)

- **Classification** (timm-style): AdamW, lr $1 \times 10^{-3} \times \frac{\text{batch}}{256}$, weight decay 0.05, cosine schedule, 5 warm-up epochs, 100 epochs, label smoothing 0.1, RandAugment (m9,n2), Mixup 0.2 / CutMix 1.0, random-resized-crop (scale 0.35–1), horizontal + vertical flip, colour jitter 0.3, EMA 0.9998, AMP, batch 64–128 at 224 px. Early stop on val macro-F1, patience 15.
- **Detection** (Ultralytics defaults): SGD lr0 0.01, momentum 0.937, wd 5×10^{-4} , warm-up 3 epochs, cosine, 200 epochs, mosaic 1.0 (closed for the last 10 epochs), mixup 0.1, HSV jitter, flips, scale 0.5, imgsz 640, batch 16–32, EMA, AMP. Early stop on val mAP@[0.5:0.95], patience 30.
- **SSL** (DINO, ViT-free version on the CNN trunk, as in solo-learn): 300 epochs on the pool, multi-crop $2 \times 224 + 6 \times 96$, AdamW, batch 128, teacher momentum 0.996→1. Alternative if compute-bound: MoCo-v3 / SimCLR 200 epochs.

11.6 Compute budget (single 12 GB GPU, e.g. RTX 3060/4070)

Item	Runs	GPU-h / run (approx.)
E1 classification: 16 models $\times$ 4 datasets $\times$ 5 seeds	320	0.5–1.5
E2 detection: 12 models $\times$ 3 datasets $\times$ 3 seeds	108	2–6
E3 cross-domain (evaluation only)	–	–
E4 SSL pre-training + 5 label fractions $\times$ 2 tasks $\times$ 3 seeds	1 + 30	40 + 1
E5 anomaly (feature extraction + memory bank)	10	0.5
E6 XAI	–	–
E8 ablations: 8 $\times$ 2 tasks $\times$ 3 seeds	48	1–4
Total		≈ 900 GPU-h ≈ 6 weeks

If this is too much, cut seeds to 3 for E1/E2 and drop the “m” scale; that halves it.

11.7 Timeline (indicative)

1. Weeks 1–2: download, deduplicate, harmonise labels, publish split files; write dataset card and start box annotation of the Custom set.
2. Weeks 3–4: implement TOMABLOCK + both YAMLS in Ultralytics; unit tests; param/FLOP table.
3. Weeks 5–8: E1, E2 (baselines run in the background while annotation continues).
4. Weeks 9–10: E4 SSL pre-training and label-efficiency.
5. Weeks 11–12: E3, E5, E6, E7, E8.
6. Weeks 13–14: figures, statistics, writing; dataset release.

12 Risks and Mitigations

Risk	Mitigation
TOMANET does not beat YOLO11/12 or ConvNeXt in-domain.	Likely on PlantVillage (saturated). The paper’s claim is <i>Pareto</i> (accuracy at ≤ 3 M params) and <i>cross-domain</i> robustness, not in-domain SOTA. Frame it that way from the start.
Custom-set annotation is slow / inconsistent.	Start now; annotate boxes only for lesion clusters; use YOLOv8 pre-labels + human correction in CVAT; measure agreement on a 100-image overlap.
SSL gives no gain over ImageNet init.	Report it honestly – a negative label-efficiency result on a 50k-image pool is itself informative; the anomaly-detection track still uses the features.
Grad-CAM metrics need masks that do not exist. Compute.	300 polygon annotations $\approx$ 2 person-days; alternatively use PlantSeg tomato masks if licence permits. Scale “n” only for baselines-vs-ours full grid; “s” only for headline datasets.
Public datasets have licence restrictions (e.g. Kaggle re-uploads without licence).	Use original sources (PlantVillage GitHub / Mendeley / Zenodo); record licence per dataset in the card.

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